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Capstone Project Report

On

PLC-Based Automatic Paint Spraying System

Submitted in partial fulfillment of the requirements
*of the **Third Year** of*
Bachelor of Technology
In
Electronics and Computer Science

by

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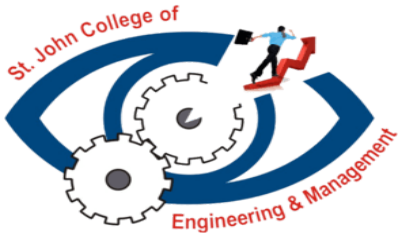
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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI

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ABSTRACT

This project presents a PLC-based automatic paint spraying system designed to automate the painting process in an industrial environment. The main objective of the system is to spray paint for 25 seconds when 15 objects have been detected inside the painting chamber. A proximity sensor is used to detect the objects moving on the conveyor, while a counter keeps track of the number of objects entering the chamber. When the counter reaches 15, the PLC stops the conveyor motor and activates the paint spray mechanism for 25 seconds. During this period, the conveyor remains stopped to ensure that the objects receive sufficient and uniform paint coverage. A timer controls the spraying duration, and after the 25-second interval is completed, the paint spray turns off and the conveyor restarts for the next batch of objects. Process Start and Process Stop switches are provided to enable and disable the system. The PLC acts as the central controller and executes the sequence through ladder logic. This automation reduces manual intervention, improves process consistency, and provides better control over the painting cycle. The system can be applied in manufacturing industries where objects require controlled and repeated paint spraying.

CONTENTS

Chapter No.	Topic	Page No.
Chapter 1	**Overview**	**1**
	1.1 Introduction	**1**
	1.2 Background	**1**
	1.3 Importance of the Project	**2**
	1.4 Perspective of Stakeholders and Customers	**3**
	1.5 Objectives and Scope of the Project	**3**
Chapter 2	**Literature Survey & Proposed Work**	**4**
	2.1 Introduction	**4**
	2.2 Literature Survey Table	**4**
	2.3 Problem Definition	**5**
	2.4 Feasibility Study	**5**
	2.5 Methodology Used	**6**
Chapter 3	**Analysis and Planning**	**7**
	3.1 Introduction	**7**
	3.2 Project Planning	**7**
Chapter 4	**Design & Implementation**	**8**
	4.1 Block Diagram	**8**
	4.2 Flowchart	**9**
	4.3 PLC I/O Assignment and Ladder Diagram	**10-11**
Chapter 5	**Results & Discussion**	**12**
	5.1 Actual Results	**12-13**
	5.2 Future Scope	**13**
Chapter 6	**Conclusion**	**14**
	References	**15**

List of Figures

Figure No.	Figure Name	Page No.
1	Gantt Chart	7
2	Block Diagram	8
3	Flow Chart	9
4	Plc Ladder Diagram	11

CHAPTER 1: OVERVIEW

1.1 Introduction

Industrial automation plays an important role in improving productivity, quality, and efficiency in manufacturing industries. Painting is a common industrial process used to protect objects from corrosion and improve their appearance. In a manual painting process, the quality and thickness of the paint may vary depending on the operator, and the process can require considerable time and effort. The PLC-Based Automatic Paint Spraying System automates the painting sequence using a Programmable Logic Controller (PLC). A proximity sensor detects objects on the conveyor, and a counter counts the objects entering the painting chamber. When 15 objects have been detected, the conveyor stops and the paint spray operates for 25 seconds. The conveyor remains stopped during spraying to prevent the objects from moving while paint is being applied. After the timer completes the 25-second interval, the spray is switched off and the conveyor restarts. Process Start and Process Stop switches allow the operator to control the system.

1.2 Background

Traditional painting processes often require manual operation, which can lead to inconsistent paint application and increased processing time. PLC-based automation provides a reliable method to control industrial painting operations. In this project, a proximity sensor detects objects on the conveyor, and a counter counts up to 15 objects. When the count reaches 15, the conveyor stops and the paint spray operates for 25 seconds using a timer. After spraying is completed, the conveyor restarts automatically. This system reduces manual intervention and improves process consistency.

1.3 Importance of the Project

The project is important because it demonstrates how PLC-based automation can improve the control of an industrial painting process.

- Automation: The system automatically controls the conveyor and paint spray according to the programmed sequence.
- Accurate Object Counting: The proximity sensor and counter help identify when 15 objects have been detected.
- Controlled Spraying Duration: The timer ensures that the paint spray operates for 25 seconds per cycle.
- Conveyor Interlocking: The conveyor stops whenever the paint spray is operating, preventing movement during the spraying period.
- Reduced Manual Intervention: The system performs the counting, stopping, spraying, and restarting operations automatically.
- Improved Consistency: A fixed spraying duration helps maintain a repeatable painting process, although actual coating quality also depends on spray pressure, paint flow, and object positioning.
- Industrial Application: The same control concept can be adapted to painting, coating, and other batch-processing operations.

1.4 Perspective of Stakeholders and Customers

- **Manufacturing Industries:** Industries can use this system to automate repetitive painting operations and improve production consistency.
- **Machine Operators:** Operators can start and stop the process using dedicated switches instead of manually controlling each stage.
- **Maintenance Personnel:** The clearly defined inputs, outputs, counter, and timer make the system easier to monitor and troubleshoot.
- **Production Supervisors:** Automatic counting and timed spraying can help maintain a predictable operating sequence.
- **Customers:** A more consistent painting process can contribute to better product appearance and finish, provided the other painting parameters are controlled appropriately.

1.5 Objectives and Scope of the Project

The main objective of this project is to design and implement a PLC-based automatic paint spraying system that controls the painting process efficiently and reliably. The system uses a proximity sensor to detect objects moving on the conveyor and a counter to count 15 objects entering the painting chamber. Once the counter reaches 15, the PLC automatically stops the conveyor motor and activates the paint spray for 25 seconds using a timer. During the spraying operation, the conveyor remains stopped to ensure that the objects do not move and receive consistent paint coverage. After the completion of the 25-second spraying cycle, the paint spray switches OFF and the conveyor restarts to process the next batch of objects. Process Start and Process Stop switches are provided to enable and disable the system. The scope of this project includes object detection, automatic counting, conveyor control, timed paint spraying, and sequential operation using PLC ladder logic.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

Industrial automation systems use sensors, controllers, motors, and actuators to perform repetitive operations with minimal human intervention. PLCs are widely used in manufacturing because they can process input signals and control output devices according to programmed logic. Timers and counters are especially useful in applications where an operation must occur after a particular number of objects have been detected or must continue for a fixed duration. In a paint spraying process, controlling the movement of objects and the spraying duration is important. The conveyor must remain stationary while paint is being applied, and the spray must stop after the specified time. The proposed system uses PLC-based sequential control to perform these operations automatically.

2.2 Literature Survey Table

Sr. No.	Research Paper Title	Author(s)	Year	Relevance / Research Gap
1	*Model Design and Simulation of Automatic Sorting Machine Using Proximity Sensor*	Bankole I. Oladapo, V. A. Balogun, A. O. M. Adeoye, C. O. Ijagbemi, Afolabi S. Oluwole, I. A. Daniyan, A. Esoso Aghor, and Asanta P. Simeon	2016	Uses a proximity sensor and PLC for object detection and conveyor control. It focuses on sorting, not painting.
2	*Design of Six Axis Automatic Painting Machine for Automatic Up and Down*	Xu-deng Xu, Mian-hao Zhang, Zhi-qiang Liu, Ze-hua Wang, and Liu-lin Wu	2017	Automates painting using PLCs and conveyor mechanisms. It focuses on multi-axis painting.
3	*Design and Fabrication of PLC Based Conveyor System with Programmable Station*	Manoj M. Nehete and Uday C. Patil	2017	Uses PLC timers and counters for conveyor control. It does not include timed paint spraying.
4	*Size Based Counting of Moving Objects on Conveyor Belt Using PLC*	Neeta Thune, Archana Kanwade, Hemlata Jadhav, Prachiti Doshi, Arya Gherde, Sarthak Jagtap, and Saumya Kudgunti	2025	Uses a PLC and vision camera for object counting. It does not focus on paint spraying.
5	*PLC-Based Automatic Paint Spraying System*	Proposed by the project team	—	Counts 15 objects, stops the conveyor, and sprays paint for 25 seconds.

2.3 Problem Definition

Manual or poorly coordinated painting systems may require operators to count objects, stop the conveyor, control the spraying duration, and restart the process. This can lead to inconsistent operation, incorrect counting, unnecessary delays, and variations in paint application. The main problem addressed by this project is the automatic coordination of object counting, conveyor movement, and paint spraying. The system must detect 15 objects, stop the conveyor, activate the paint spray for 25 seconds, and ensure that the conveyor remains stopped throughout the spraying period. The spray must turn off when the timer finishes, after which the conveyor can restart for the next batch. Process Start and Process Stop switches are provided to control the overall operation.

2.4 Feasibility Study

Technical Feasibility: The project can be implemented using a PLC with suitable digital inputs and outputs, a proximity sensor, a conveyor motor control circuit, a paint spray control device, a timer, and a counter. Ladder logic can coordinate the operation of all these components.

Operational Feasibility: The system is simple to operate. The operator presses the Process Start switch to enable automatic operation and the Process Stop switch to disable it. Object counting, conveyor stopping, spraying, and restarting are controlled automatically.

Economic Feasibility: Using standard industrial automation components can reduce repetitive manual work and improve process consistency. The actual cost depends on the PLC model, motor rating, sensor type, spray equipment, and required safety equipment.

Safety Feasibility: Since the system involves paint spraying, the actual installation must include suitable electrical protection, ventilation, and equipment rated for the paint and surrounding environment. The Process Stop switch must not be treated as a substitute for an emergency-stop circuit where one is required.

2.5 Methodology Used

The methodology used in this project involves automating the paint spraying process using a Programmable Logic Controller (PLC). Initially, the system is enabled by pressing the Process Start switch (X2), and the conveyor motor (Y1) starts moving the objects towards the painting chamber. The proximity sensor (X0) detects each object, and the PLC counter (C0) counts the objects until the preset value of 15 is reached. Once 15 objects are detected, the PLC stops the conveyor motor and activates the paint spray (Y0). Timer T0 controls the spraying duration for 25 seconds while the conveyor remains stopped. After the timer completes, the paint spray switches OFF and the conveyor restarts for the next batch. The Process Stop switch (X1) is used to disable the system and stop its operation. This sequence ensures automatic object counting, controlled spraying time, and proper coordination between the conveyor and paint spray.

Main steps involved:

1. Start the system and detect objects using the proximity sensor.
2. Count 15 objects and stop the conveyor.
3. Operate the paint spray for 25 seconds while the conveyor remains stopped.
4. Turn OFF the spray and restart the conveyor for the next batch.

CHAPTER 3 : ANALYSIS AND PLANNING

3.1 Introduction

The analysis and planning stage identifies the components, input/output assignments, operating sequence, and control conditions required for the project. The system must correctly count 15 objects, stop the conveyor during paint spraying, and maintain the spray operation for exactly the programmed 25-second interval. The PLC acts as the main controller. The proximity sensor supplies object detection signals, the counter monitors the number of objects, and the timer controls the spraying duration. The start and stop switches allow the operator to enable or disable the process. Proper planning of the PLC ladder logic is necessary to prevent the conveyor and paint spray from operating simultaneously.

3.2 Project Planning

GANTT CHART

PLC-Based Automatic Paint Spraying System

Phase	Activity	Week							
		1	2	3	4	5	6	7	8
1. Requirement Analysis	Study problem statement								
	Define objectives and scope								
2. Component Selection	Select PLC, sensors, motor, spray								
	Finalize hardware								
3. Design & Planning	Create block diagram, DFD, flow chart								
	Prepare I/O table and ladder logic								
4. Development & Implementation	Write and simulate PLC program								
	Integrate hardware and test								
5. Documentation	Prepare report								
	Finalize and submit								

CHAPTER 4 : DESIGN & IMPLEMENTATION

4.1 Block Diagram

The block diagram represents the working of the PLC-based automatic paint spraying system. The proximity sensor (X0), Process Start switch (X2), and Process Stop switch (X1) provide inputs to the PLC. The PLC processes these inputs using counter C0 and timer T0. When 15 objects are detected, the PLC stops the conveyor motor (Y1) and activates the paint spray (Y0) for 25 seconds. After spraying is completed, the spray switches OFF and the conveyor restarts for the next batch. This arrangement ensures automatic object counting, timed spraying, and coordinated system operation.

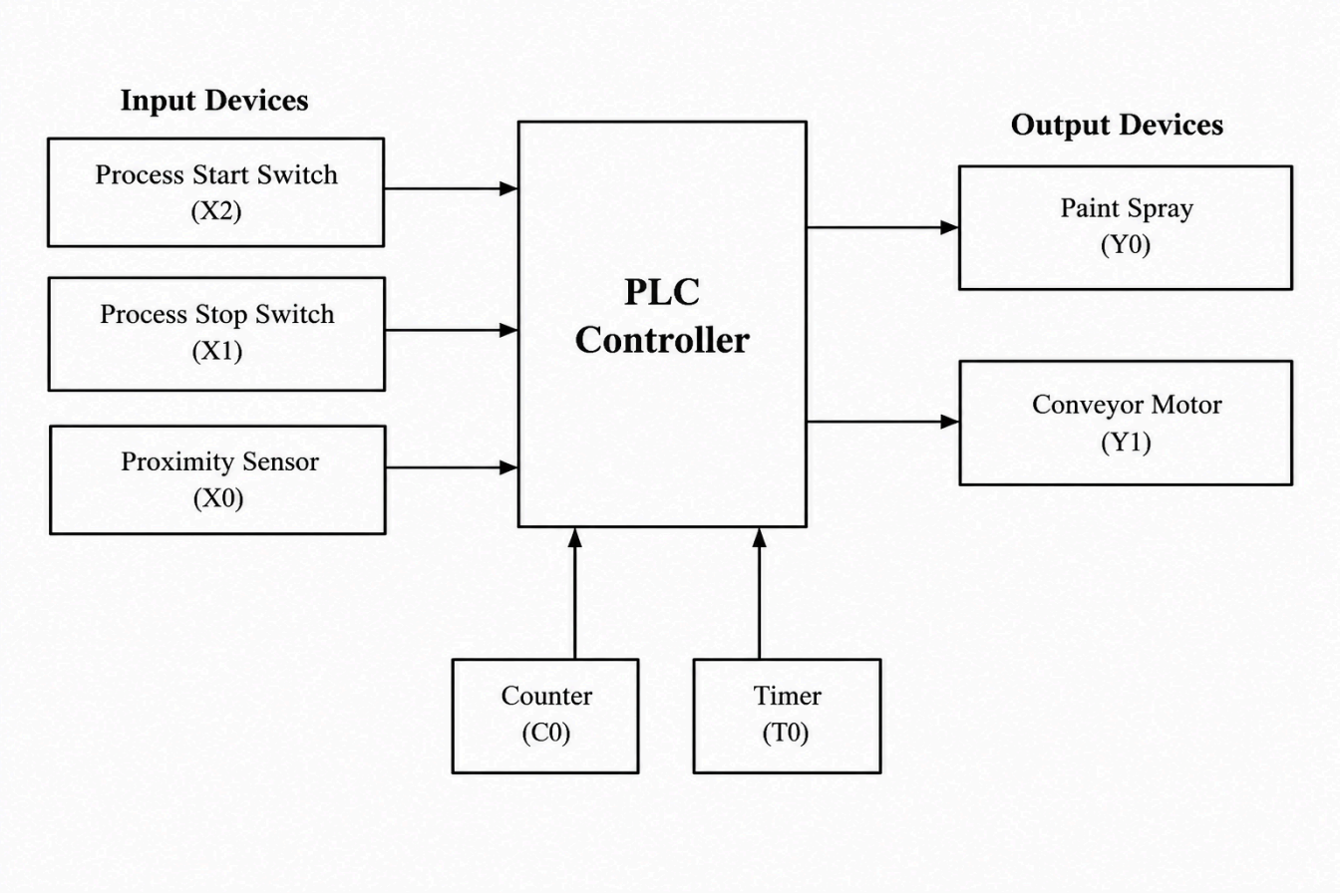


Figure 4.1

4.2 Flowchart

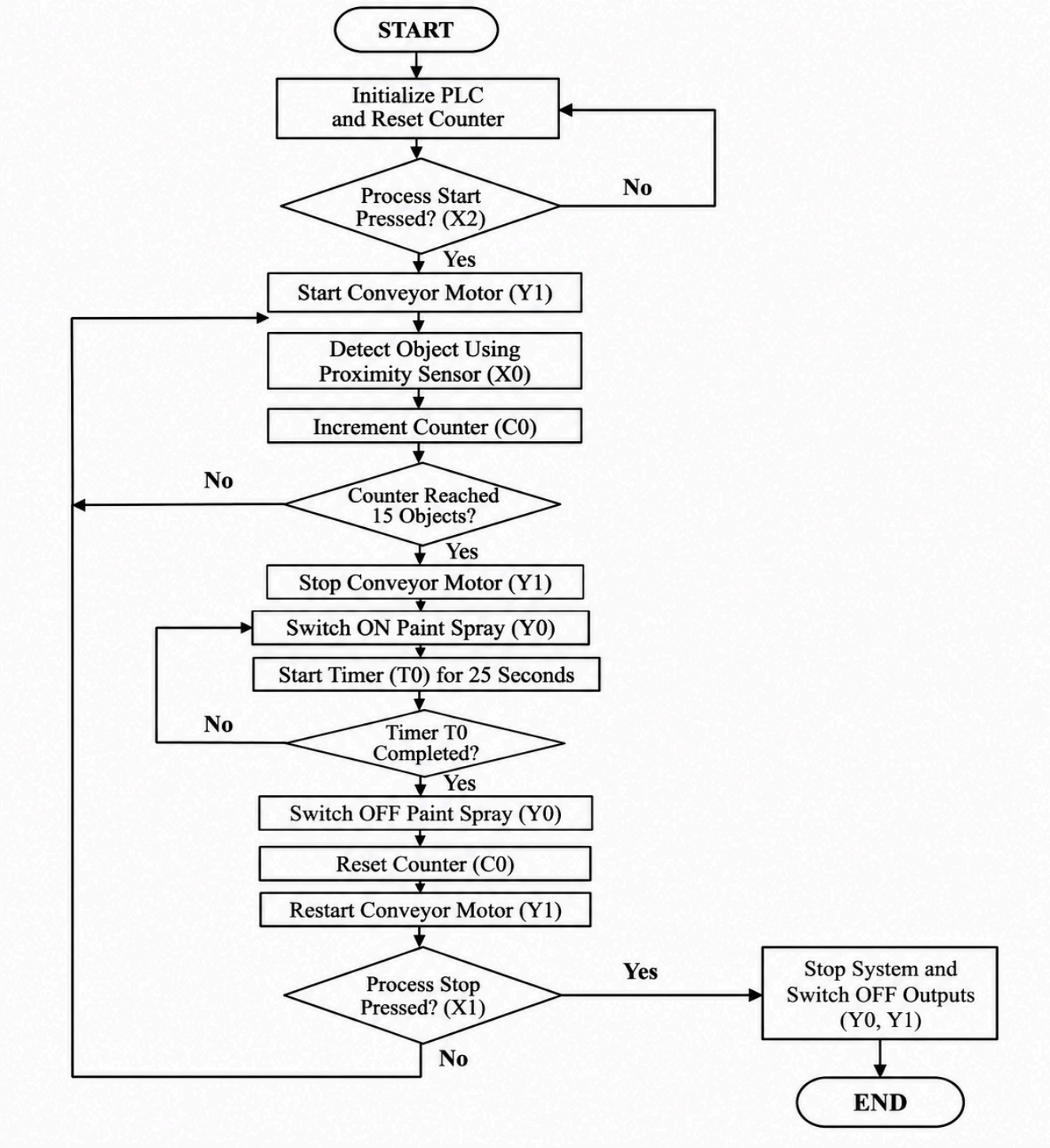


Figure 4.2

4.3 PLC I/O Assignment and Ladder Diagram

4.3.1 PLC I/O Assignment Table

Device	Type	Function / Description
X0	Digital Input	Proximity sensor for object detection
X1	Digital Input	Process Stop switch
X2	Digital Input	Process Start switch
Y0	Digital Output	Paint spray control
Y1	Digital Output	Conveyor motor control
T0	Timer	Controls paint spraying duration of 25 seconds
C0	Counter	Counts 15 objects

4.3.2 Working of PLC Ladder Logic

The PLC ladder logic controls the automatic paint spraying process. When the Process Start switch (X2) is pressed, the conveyor motor (Y1) starts. The proximity sensor (X0) detects and counts objects using counter C0. When 15 objects are counted, the conveyor stops and the paint spray (Y0) operates for 25 seconds using timer T0. After spraying is completed, the spray turns OFF and the conveyor restarts. The Process Stop switch (X1) disables the system.

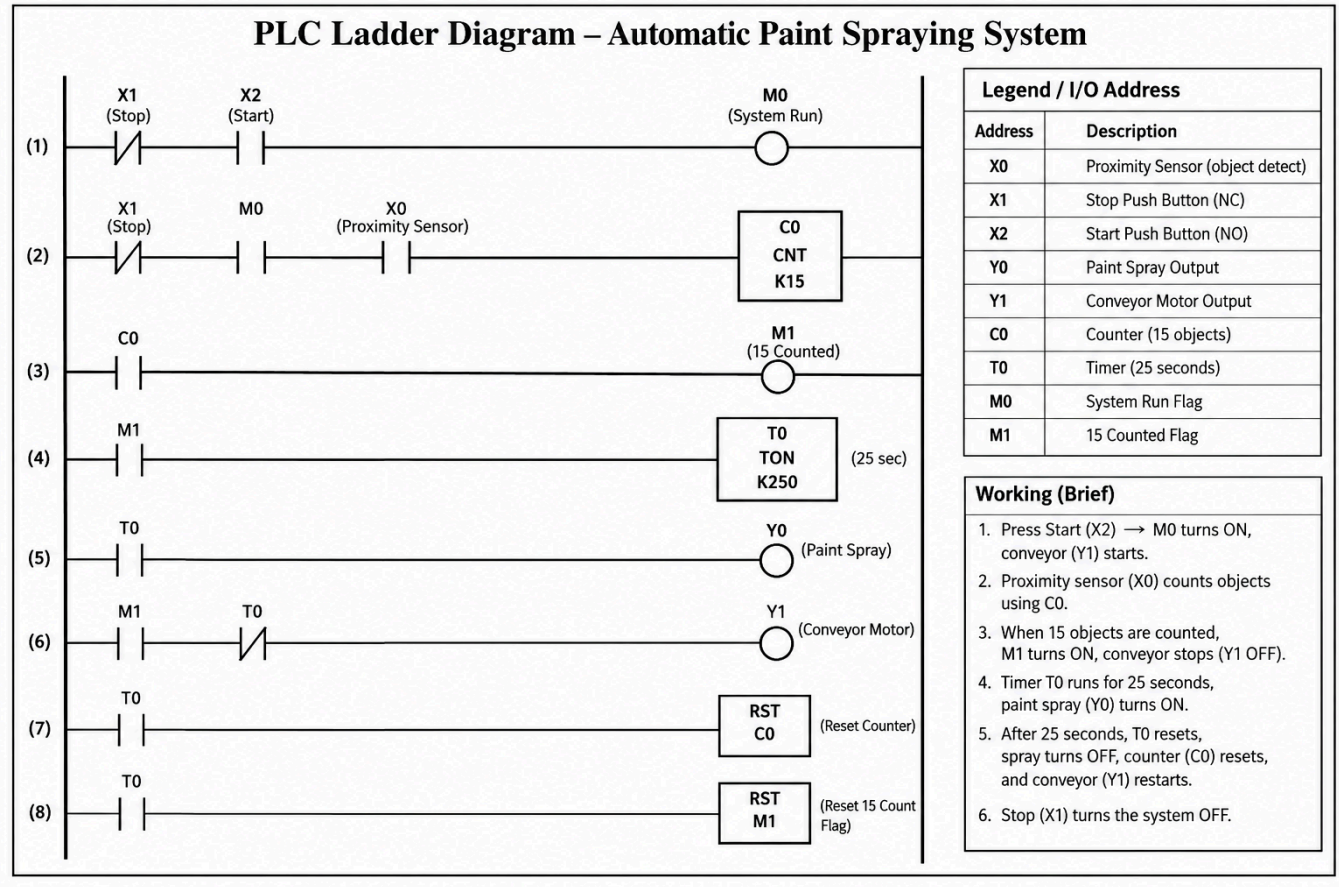
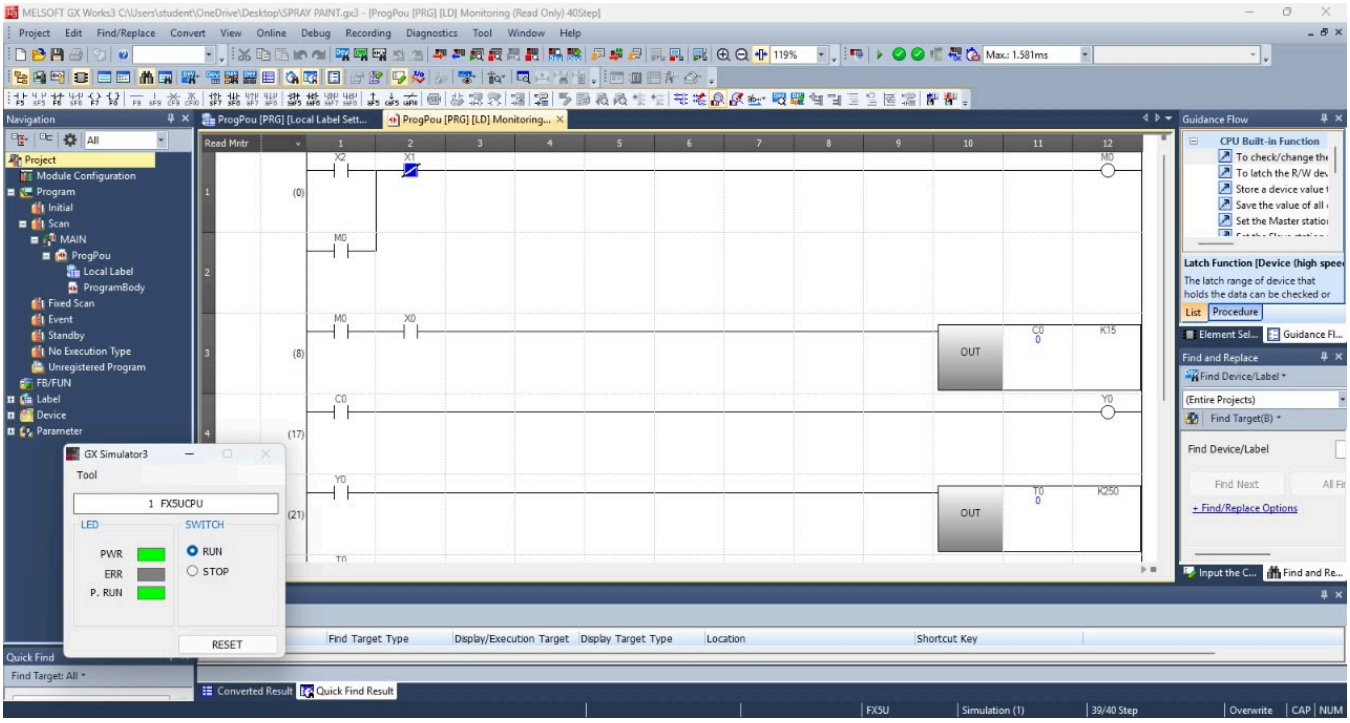


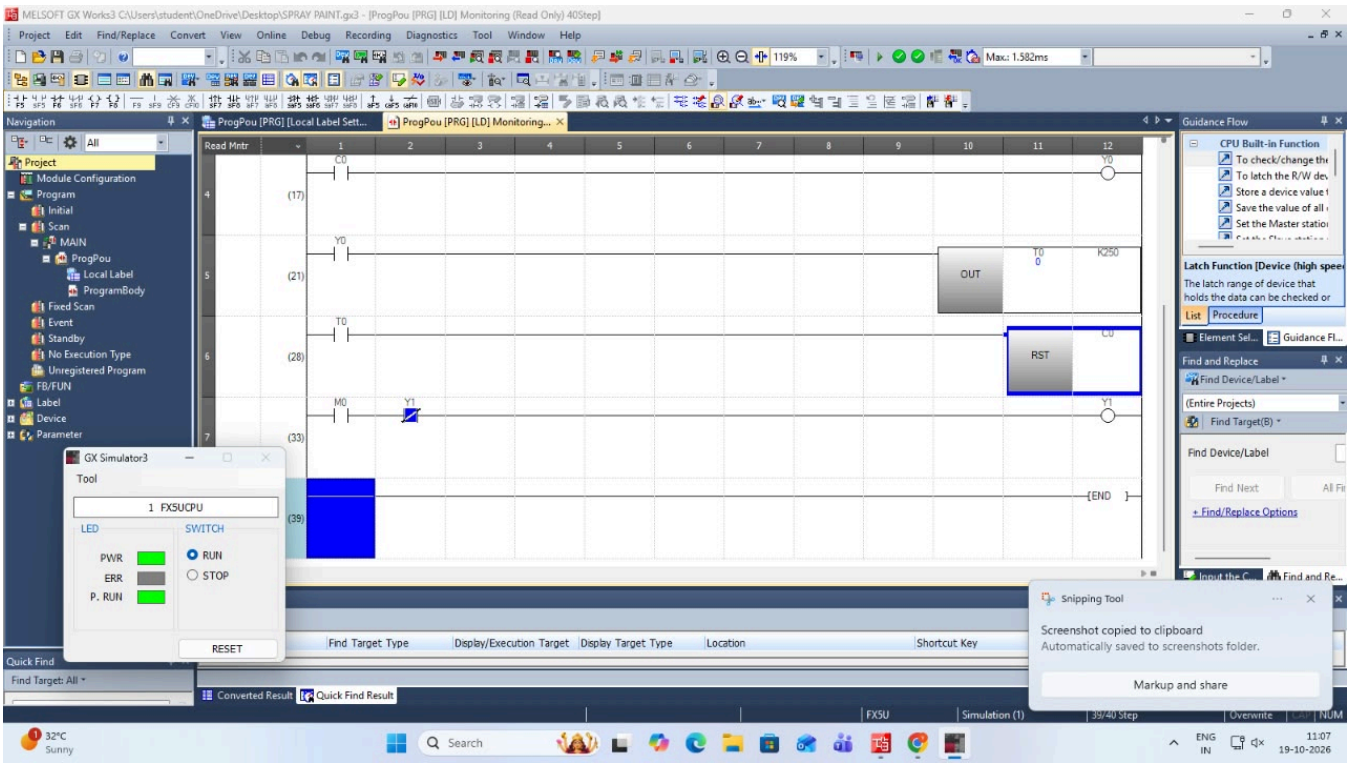
Figure 4.3.2

CHAPTER 5 : RESULTS & DISCUSSION

5.1 Actual Results

The PLC-based automatic paint spraying system performed the object-counting and painting sequence according to the programmed logic. The proximity sensor (X0) detected the objects on the conveyor, and the counter (C0) counted them until the preset value of 15 was reached. At this count, the conveyor motor (Y1) switched OFF, and the paint spray (Y0) switched ON for 25 seconds, controlled by timer (T0). During spraying, the conveyor remained stopped. After the timer completed, the paint spray switched OFF and the conveyor restarted for the next batch. The Process Start switch (X2) enabled the system, while the Process Stop switch (X1) stopped its operation. Thus, the programmed sequence demonstrated automatic object counting, timed spraying, and coordinated conveyor control.





5.2 Future Scope

The project can be extended with the following features:

1. HMI Integration: Display object count, timer, and system status.
2. Adjustable Batch Size: Change the number of objects per batch.
3. Paint-Level Monitoring: Detect low paint levels and alert the operator.
4. Safety Interlocks: Add emergency stops and safety-door switches.
5. Spray Quality Monitoring: Monitor paint pressure and flow.
6. Data Logging: Record batch counts, cycle times, and faults.
7. SCADA Integration: Enable centralized monitoring and control.

CHAPTER 6 : CONCLUSION

The PLC-based automatic paint spraying system performed the object-counting and painting sequence according to the programmed logic. The proximity sensor (X0) detected the objects on the conveyor, and the counter (C0) counted them until the preset value of 15 was reached. At this count, the conveyor motor (Y1) switched OFF, and the paint spray (Y0) switched ON for 25 seconds, controlled by timer (T0). During spraying, the conveyor remained stopped. After the timer completed, the paint spray switched OFF and the conveyor restarted for the next batch. The Process Start switch (X2) enabled the system, while the Process Stop switch (X1) stopped its operation. Thus, the programmed sequence demonstrated automatic object counting, timed spraying, and coordinated conveyor control.

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